Project Proposal

E-voting system

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Abstract

The AI-Based Secure E-Voting System is a web-based platform designed to enhance the integrity, security, and efficiency of electoral processes through Artificial Intelligence and Machine Learning. The system integrates AI-based face recognition for voter authentication and ML-based fraud detection to prevent duplicate or fraudulent votes. By ensuring secure, transparent, and tamper-proof voting, this system aims to modernize traditional voting methods and build trust in digital democracy.

Background of the Project

Voting is a cornerstone of democratic systems, yet traditional methods are often slow, prone to human error, and vulnerable to fraud. With the rise of digital technologies, there is a growing opportunity to develop secure, efficient, and accessible electronic voting systems. This project leverages advancements in AI and ML to create a robust e-voting platform that addresses key challenges in voter authentication, vote integrity, and result transparency.

Problem Statement

Current voting systems face several critical issues:

- Risk of unauthorized voting using fake identities.
- Possibility of duplicate voting by the same individual.
- Vulnerability to tampering and unauthorized data access.
- Lack of transparency leading to low user trust.
- Slow and error-prone manual vote counting.

Solution Overview

The proposed system utilizes AI-based face recognition for secure voter authentication and machine learning algorithms to detect and prevent fraudulent activities. It ensures one voter = one vote through identity verification and database checks. The platform also provides encrypted data storage, Realtime monitoring, and automated result generation to enhance security, transparency, and efficiency.

Objective of the project

General Objective:
To develop a secure, reliable, and user-friendly online voting system powered by AI and ML technologies.

Specific Objectives:

- To implement AI-based face recognition for voter identity verification.
- To prevent fake and duplicate voting using ML-based fraud detection.
- To ensure each voter can cast only one vote.
- To provide accurate, transparent, and tamper-proof election results.
- To create an intuitive and accessible voting interface.

Scope of the project

This project focuses on developing a web-based e-voting system for organizations, institutions, or small-scale elections. It includes voter registration, authentication, voting, real-time monitoring, and result declaration modules. Future enhancements may include blockchain integration, mobile app support, and multi-language interfaces.

Methodology

The project follows the **Waterfall Methodology**, ensuring structured and documented development across the following phases:

1. Requirement Analysis and Definition
2. System and Software Design
3. Implementation and Unit Testing
4. Integration and System Testing
5. Deployment and Maintenance

Reason for Chosen Methodology

The Waterfall model is suitable for this project due to its clear phase-wise progression, thorough documentation, and client approval checkpoints, ensuring alignment with security and functional requirements.

Development Tools

Activities	Tools/Programs
Frontend development	HTML, CSS, JavaScript
Backend development	Python (Flask)
Database	MySQL
Platform	Windows
AI/ML Integration	OpenCV, Face Recognition, Scikit-learn
Browsers	Google chrome
Documentation	MS Word
Diagram	Draw.io

Table1

Expected Outcomes

- A fully functional, secure, and scalable e-voting platform.
- Prevention of duplicate and fraudulent voting attempts.
- Real-time fraud detection and activity monitoring.
- Automated and accurate vote counting.
- Enhanced voter trust through transparency and security.

Conclusion

The AI-Based Secure E-Voting System represents a significant step toward modernizing electoral processes through technology. By integrating AI and ML, the system ensures security, accuracy, and transparency, making it a reliable alternative to traditional voting methods. This project aims to contribute to the evolution of digital democracy and trusted electoral systems.